Softmax Regression Studied from First Principles

ML Assignment A2 — Option 1: Study a fundamental machine-learning model
Student: <STUDENT NAME> · ID: <STUDENT ID> · UTS, <YEAR> · File name: `STUDENTNAME_STUDENTID_YEAR_UTS_ML_Journal.pdf`

1. Link to implementation

Public Colab notebook (plain-text URL): **<PASTE PUBLIC COLAB URL>**
Archived copy: `https://github.com/yeamin1414/advanced-analytics/blob/claude/busy-bardeen-sep7tp/A2_Option1_Softmax_Regression.ipynb`
The notebook is self-contained (downloads Fashion-MNIST, pre-processes, trains, evaluates, saves figures and `results_opt1.json`), seeded (`SEED=2025`) and runs in ≈ 10–25 min on CPU. Softmax regression, both losses, all gradients, GD / momentum / Adam / L-BFGS, cross-validation and all metrics are implemented in NumPy; scikit-learn/SciPy are used only in cells labelled *verification* or *reference*.

2. Project report

2.1 Problem definition

Task. Supervised 10-class classification of garment images (Fashion-MNIST), used as a vehicle to *study* the softmax-regression model: how faithfully does minimising the convex cross-entropy surrogate track the quantities we actually care about (0-1 error, cost-weighted error, calibration), and which mechanisms control the gaps?

	Training phase	Deployment phase
Input	$X \in \mathbb{R}^{n\times 784}$, n = 50,000. Row = a 28×28 greyscale catalogue image flattened row-major, raw integers 0…255 (0 = background). Pre-processing $\tilde{x} = (x/255 - \mu)/\sigma$ with $\mu \in \mathbb{R}^{784}$ the per-pixel *training* mean and $\sigma = 0.295$ one global *training* std (centred, front-view, background-removed product photos; not photos in the wild).	One 28×28 uint8 image through the same frozen (μ, σ). Out of scope: colour, rotated, cluttered images.
Output / label	$y \in \{0..9\}$ (T-shirt, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot) as one-hot e_y; ≈ 5,000 per class.	Probability vector $q \in \Delta^9$ ($q_k \geq 0$, $\Sigma q_k = 1$) and the decision $\hat{y} = \text{argmax}_k\, q_k$ (or the minimum-expected-cost class).
Compared via	q vs e_y through cross-entropy (convex surrogate).	$\hat{y}$ vs truth through the **task objective**: 0-1 error / cost-weighted error; q through calibration (ECE).

Splits. 60,000 official training images → 50,000 train / 10,000 validation (seeded random split); the official 10,000 test images are used once, after all choices are frozen.

Challenges. (i) Four upper-body classes (T-shirt/Shirt/Pullover/Coat) are visually near-identical, so a linear model cannot reach low error; (ii) pixel correlation ⇒ ill-conditioned

optimisation ($\kappa \le 1.1e{+}04$); (iii) softmax is over-parameterised (shift invariance) ⇒ convex but not strictly convex; (iv) the training loss is not the task objective.

Motivation and historical perspective

Softmax regression is the simplest model in which every ingredient of modern ML is visible and analysable: a hypothesis space, an information-theoretic loss, a convex optimisation problem, a norm-based generalisation bound, and a probabilistic output. The last layer of nearly every deep classifier *is* a softmax regression, so understanding it exactly transfers to deep networks. **History:** Berkson (1944) introduced the logit; Cox (1958) formalised logistic regression; the multinomial form is Theil (1969) / McFadden (1974); Rosenblatt's perceptron (1958) gave a linear classifier without probabilities; Jaynes' maximum-entropy principle (1957) shows softmax regression is exactly the max-entropy distribution matching empirical feature moments (Berger et al. 1996 for NLP); Bridle (1990) named “softmax”. Learning theory then asked why minimising a convex *surrogate* is legitimate: Zhang (2004) and Bartlett–Jordan–McAuliffe (2006) proved calibration/consistency of cross-entropy — but only for unrestricted function classes and infinite data; Bartlett–Mendelson (2002) and Kakade–Sridharan–Tewari (2008) gave norm-based bounds $\propto BR/\sqrt{n}$ for linear predictors; Soudry et al. (2018) showed gradient descent on separable data has an implicit bias to the max-margin solution with $\|W\|$ growing only like $\log t$; Guo et al. (2017) documented mis-calibration of CE-trained models; Zhang & Sabuncu (2018) and Ghosh et al. (2017) studied bounded losses robust to label noise. **Why ML rather than rules:** no closed-form rule maps 784 correlated intensities to “Coat” vs “Shirt”, while 60k i.i.d. labelled examples from a stable catalogue distribution are available — the conditions that justify empirical risk minimisation. **Why linear rather than a CNN:** the linear model is the object of study: convex (optimisation separable from modelling), capacity controlled by $\|W\|$ (theory testable), and its residual error quantifies what non-linear features add.

Research question and hypotheses

- **H1 (capacity):** the generalisation gap is governed by $\|W\|_F R/\sqrt{n}$, not by the number of parameters (7,850).
- **H2 (loss ≠ objective):** CE is more sensitive than 0-1 error to over-fitting and to rescaling of logits; the mismatch is measurable and repairable (temperature scaling).
- **H3 (optimisation as regulariser):** with $\lambda = 0$ on separable data the loss is never minimised: $\|W\| \propto \log t$ and the margin grows.
- **H4 (customised loss):** under symmetric label noise a bounded re-weighted CE (generalised CE) is more robust than CE at no cost on clean data.

2.2 Machine-learning approach

Model and justification

Hypothesis space: $q(x;W,b) = \mathrm{softmax}(W^T x + b)$, $W \in \mathbb{R}^{784\times 10}$, $b \in \mathbb{R}^{10}$ (7,850 parameters). Linear log-odds is the exponential-family/GLM choice, equals the max-entropy distribution with matching feature moments, and has a convex negative log-likelihood. Shift invariance $\mathrm{softmax}(z + c1) = \mathrm{softmax}(z)$ creates 785 flat directions of the unregularised loss (verified numerically below); the L2 penalty removes them.

Loss and gradient. $\ell_{CE}(z,y) = -\log q_y = H(e_y, q) = KL(e_y\|q)$ (since $H(e_y) = 0$), so ERM = maximum likelihood. $\partial\ell/\partial z = q - e_y$, hence $\nabla_W J = X^T(Q-Y)/n + \lambda W$, $\nabla_b J = 1^T(Q-Y)/n$ with $J =$ mean $\ell + (\lambda/2)\|W\|^2_F$ (bias unpenalised). At a $\lambda = 0$ optimum $X^T(Q - Y) = 0$ (moment

matching). The Hessian $(1/n)\Sigma(\mathrm{diag}\ q_i - q_i q_i^T) \otimes \tilde{x}_i \tilde{x}_i^T$ is PSD (convex) and $\preceq L \cdot I$ with $L = \tfrac{1}{2}\lambda_{max}(\tilde{X}^T\tilde{X}/n) + \lambda$, giving a principled step $1/L$ and $\kappa \le L/\lambda$.

Customised component. Generalised cross-entropy $\ell_q = (1 - q_y^q)/q$ ($\to$ CE as $q \to 0$, = MAE at $q = 1$). I derived $\partial \ell_q / \partial z_j = q_y^q (q_j - 1[j=y])$ — CE's gradient re-weighted by q_y^q, so likely-mislabelled points are down-weighted — and verified it by finite differences.

System structure and theory-to-code

```
raw 28x28 uint8 → x/255 → (x−μ)/σ → z = Wᵀx+b → q = softmax(z) → ŷ = argmax q  (or argmin_k Σ_j q_j C[j,k])
training: J(θ) = mean ℓ + λ/2‖W‖² → analytic ∇J → optimiser (GD / momentum / Adam / L-BFGS) → θ*;
validation → λ (5-fold CV), T;  test → once
```

Maths	Code (notebook)
softmax, log-softmax (log-sum-exp shift)	`softmax`, `log_softmax`
ℓ_{CE} and $(q - e_y)/n$	`ce_loss_grad`
ℓ_q and $q_y^q(q - e_y)/n$	`gce_loss_grad`
$J(\theta)$, ∇J via chain rule through $z = \tilde{X}\theta$	`make_objective`
GD, heavy-ball, Adam (bias-corrected)	`GD`, `Momentum`, `Adam`, `fit_first_order`
L-BFGS two-loop recursion + Armijo	`lbfgs`
error, macro-F1, Brier, ECE, confusion	`evaluate`, `macro_f1`, `ece_score`, `confusion`

Training procedure and hyper-parameters

Item	Setting (reason)
Pre-processing	/255, per-pixel train mean, one global train std (keeps geometry; per-pixel std would amplify near-constant border pixels)
Regularisation	L2 on W; $\lambda \in \{1e\text{-}6, \ldots, 1e\text{-}1\}$ chosen by 5-fold CV on a 20k subset; **$\lambda^* = 0.01$**
Main solver	own L-BFGS, m = 10, Armijo $c_1 = 1e\text{-}4$, up to 800 iterations on the final fit (converged solutions needed so early stopping does not confound λ)
GD / momentum	$\eta = 1/L$ ($L = 113.4$ computed from data); heavy-ball $\beta = 0.9$
Adam	$\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 1e\text{-}8$; lr 1e-2 (convergence comparison), 3e-3 (over-fitting study), 2e-3 with batch 64 (noise study)
Seeds	SEED = 2025 (split), fixed seeds per experiment; 5 folds / 5 seeds / 2 subsets for ± values

Evaluation metrics and relevance

0-1 error — the task objective (classes balanced, so accuracy is not misleading); **macro-F1 + per-class P/R/F1 + confusion matrix** — class-specific failures hidden by the mean; **cross-entropy** — the training loss, a proper scoring rule; **Brier** — second bounded proper score; **ECE** (15 bins) — whether stated confidence matches accuracy. Baselines: majority class, nearest-class-mean. Uncertainty: fold std, seed std and bootstrap 95 % CI. The ideal comparison (0-1 loss) is not used for training because it is piecewise constant (zero gradient a.e.) and NP-hard to minimise for linear classes; it is measured on validation/test instead.

2.3 Results

Implementation verification

Test	Result
Loss at θ = 0	2.302585 = ln 10
Finite-difference gradient check (max relative error)	CE 7.7e-07; GCE 1.1e-07
Analytic vs finite-difference Hessian	max |Δ| = 4.5e-11; min eigenvalue -5e-17 (PSD ⇒ convex); 6 zero eigenvalues = predicted d+1 = 6
Own L-BFGS vs scikit-learn `LogisticRegression` (same objective, λ = 1e-2, n = 20k)	|Δ objective| = 5.8e-13; max |Δ probability| = 3.9e-05; predictions agree 100%
Moment matching at λ = 0 (n = 10k)	max $\|E_{data}[x\ y_k] - E_{model}[x\ q_k]\|$ = 3.8e-04 (typical moment 4.8e-02)

Optimiser comparison (n = 20k, λ = 1e-2, 300 iterations; sub-optimality f − f*)

Method	f − f*
GD lr=1/L	1.37e-01
GD lr=1.9/L	9.01e-02
GD lr=8/L	3.03e-02
Momentum lr=1/L, mu=0.9	2.50e-02
Adam lr=1e-2	3.37e-03
Adam lr=1e-3	2.12e-02
L-BFGS (own)	1.68e-04

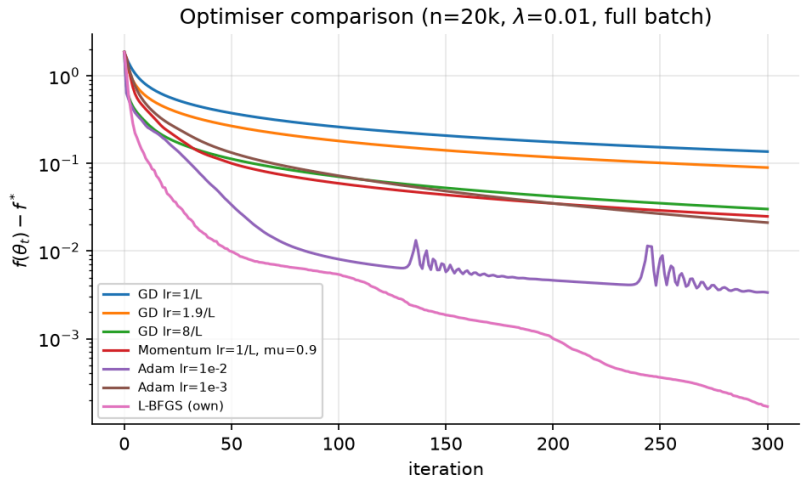


Figure 1. Convergence; κ ≤ 1.1·10⁴. GD converges even at 8/L ≫ 2/L because the ½ bound on the softmax Hessian is loose near the optimum.

Model family (random members, neighbourhood, Naive Update)

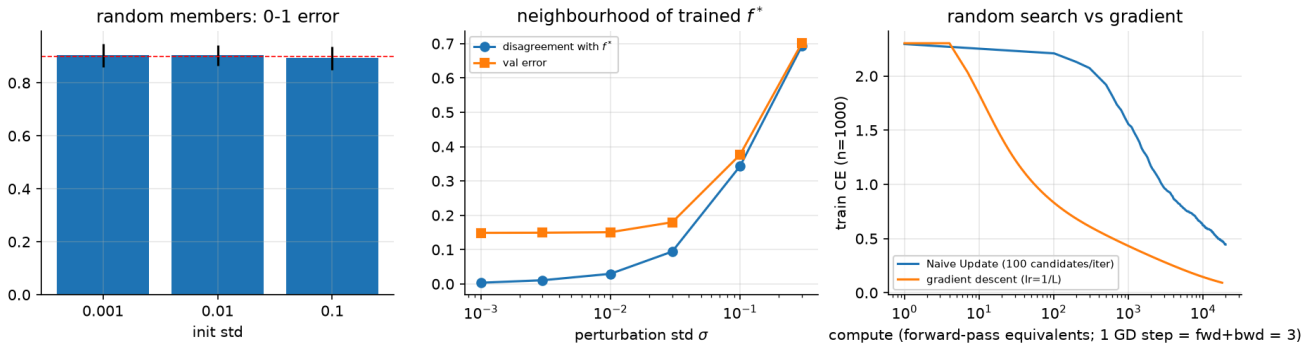


Figure 2. Left: 100 random members are at chance (error ≈ 0.90, mutual disagreement ≈ 0.90). Middle: predictions stable for σ ≲ 0.01 around f*. Right: Naive Update (CE 0.445 after 200 iterations = 20,000 forward passes) vs gradient descent: 0.505 after 200 steps, 0.0926 at equal compute (6,000 steps).

Model selection (5-fold CV, 20k subset)

λ	CV cross-entropy	CV error	train CE
1e-06	0.6813 ± 0.0403	0.1809 ± 0.0083	0.2405
1e-05	0.6655 ± 0.0280	0.1808 ± 0.0065	0.2397
0.0001	0.6159 ± 0.0345	0.1768 ± 0.0074	0.2424
0.001	0.4771 ± 0.0219	0.1605 ± 0.0076	0.2798
0.01	0.4353 ± 0.0145	0.1495 ± 0.0088	0.3630
0.1	0.5044 ± 0.0112	0.1651 ± 0.0082	0.4841

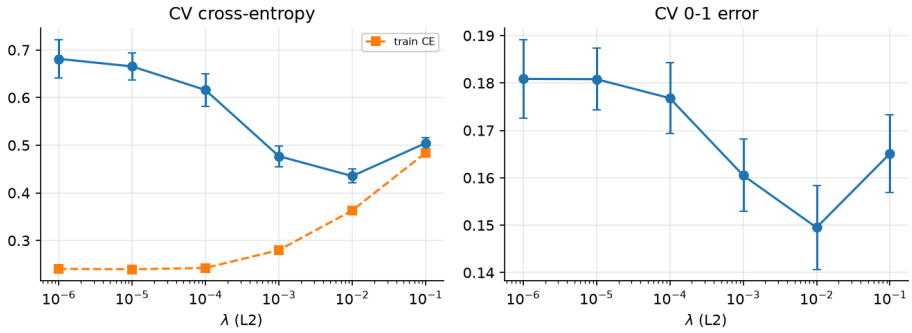


Figure 3. U-shaped CV curves; λ* = 1e-2 minimises both CE and error.

Final model (λ* = 0.01, 50k train)

Split	Error	Macro-F1	CE	Brier	ECE
train	13.5%	0.864	0.398	0.200	0.0374
val	14.7%	0.851	0.432	0.216	0.0287
test	16.0%	0.839	0.448	0.226	0.0175

Baselines (test error): majority class 90.0%; nearest-class-mean 32.3%; **softmax regression 15.95%** (95 % bootstrap CI [15.14%, 16.66%]); reference off-the-shelf MLP-256 12.1% (not my implementation; context only).

Class	Precision	Recall	F1
T-shirt	0.808	0.820	0.814
Trouser	0.979	0.951	0.965
Pullover	0.726	0.730	0.728

Dress	0.833	0.871	0.852
Coat	0.726	0.761	0.743
Sandal	0.926	0.915	0.921
Shirt	0.630	0.559	0.592
Sneaker	0.899	0.924	0.911
Bag	0.930	0.939	0.934
Ankle boot	0.934	0.935	0.935

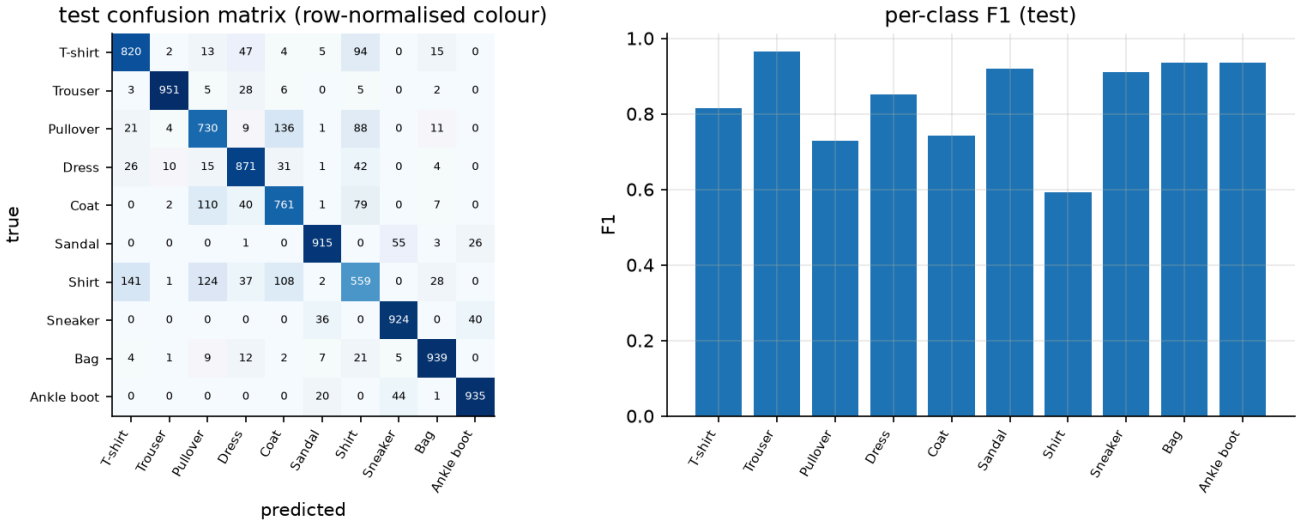


Figure 4. Test confusion matrix and per-class F1. Shirt (F1 0.59) is confused with T-shirt, Pullover and Coat; Pullover↔Coat is next.

Loss vs task objective

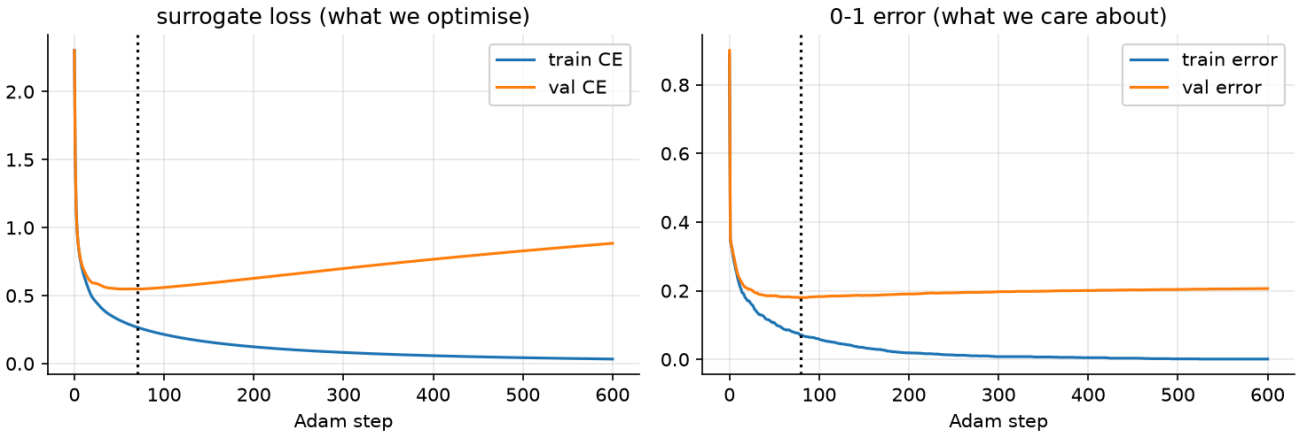


Figure 5. n = 2,000, λ = 0, Adam. Train CE and error fall monotonically; validation CE bottoms at step 70 (0.548) then rises 61.4%, whereas validation error rises only 14.7% (0.180 → 0.206).

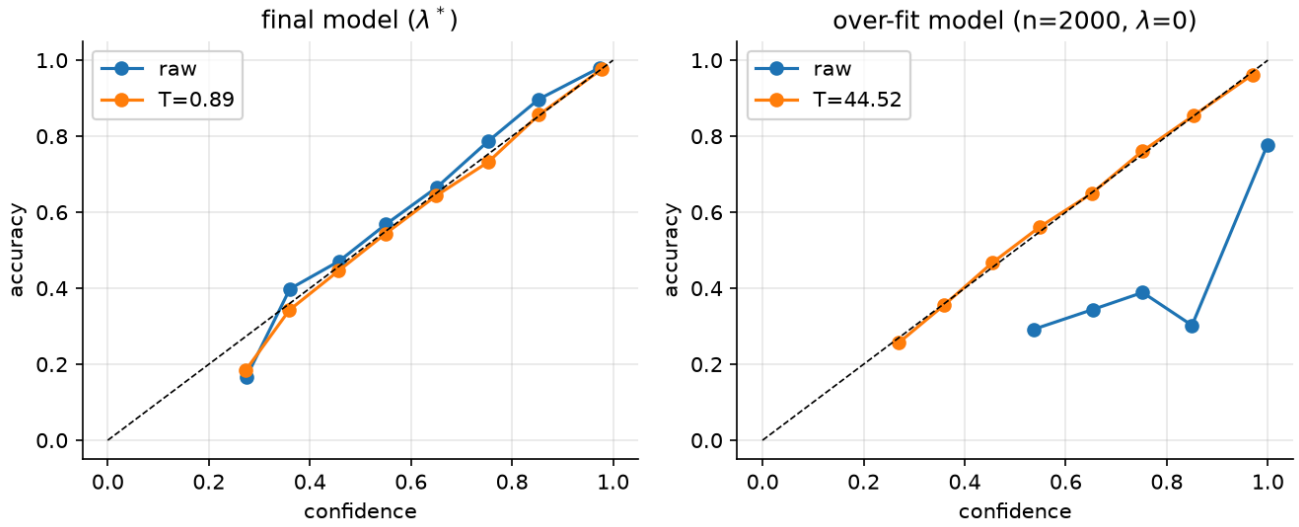


Figure 6. Reliability diagrams. Final model: T = 0.89, ECE 0.0175 → 0.0084. Over-fit model (n = 2000, λ = 0): T = 44.5, test CE 16.10 → 0.66, ECE 0.224 → 0.013, with identical error 22.8%.

The worst 5 % of test points carry 36.1% of the total test log-loss (10 % → 54.8%; 20 % → 76.5%); 27.6% of errors are near-misses ($q_{true} > 0.3$).

Implicit bias, generalisation, label noise, cost-sensitive decisions

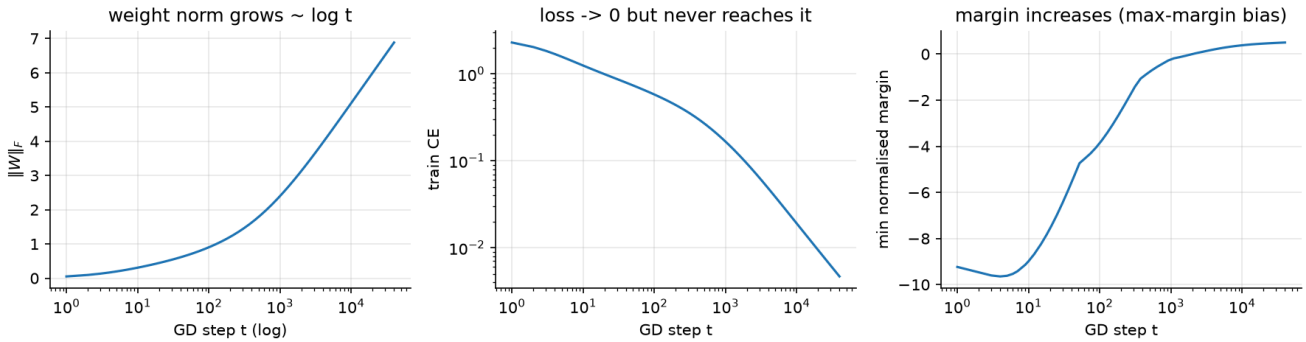


Figure 7. n = 500, λ = 0, 40,000 GD steps: $\|W\|_F = 1.24 \ln t - 6.25$ ($R^2 = 0.9994$); final train loss 0.0047; normalised margin → 0.50.

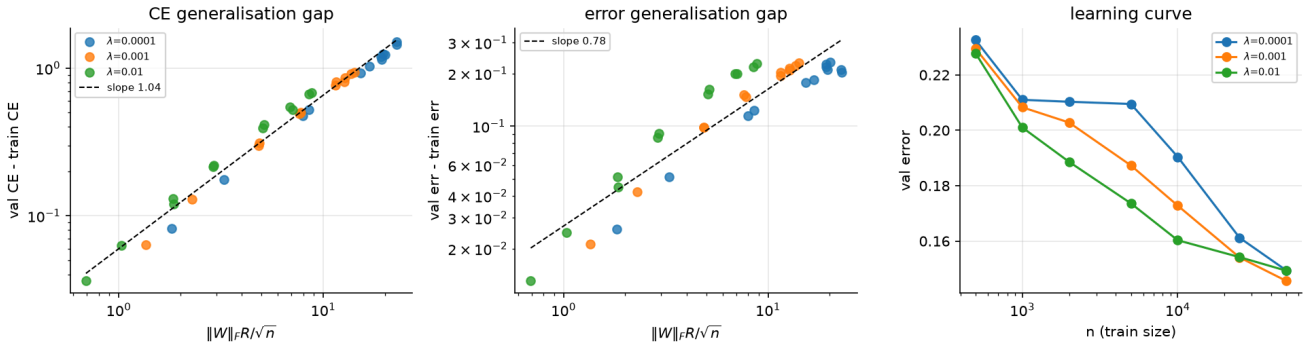


Figure 8. Generalisation gap vs $\|\hat{W}\|_F R/\sqrt{n}$ (R = 51.2): Spearman ρ = 0.989 (CE gap), 0.865 (error gap); log-log slopes 1.04 and 0.78; error gap ∝ $n^{-0.58}$. Parameter count is constant (7,850).

λ	Loss	clean test error, 0 % noise	clean test error, 40 % noise
0.001	CE	0.2009 ± 0.0039	0.4053 ± 0.0204
0.001	GCE q=0.7	0.1842 ± 0.0016	0.2267 ± 0.0033
0.001	GCE q=0.3	0.1931 ± 0.0034	0.3717 ± 0.0196

0.01	CE	0.1954 ± 0.0036	0.3492 ± 0.0187
0.01	GCE q=0.7	0.1878 ± 0.0053	0.2176 ± 0.0112
0.01	GCE q=0.3	0.1877 ± 0.0032	0.2953 ± 0.0208

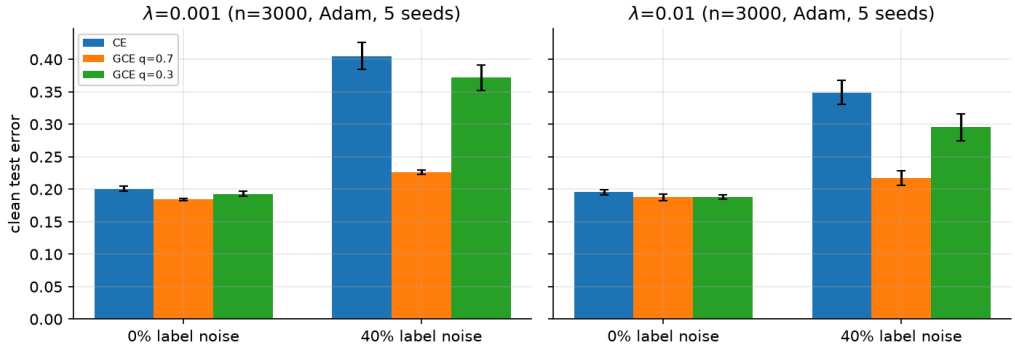


Figure 9. n = 3000, Adam, 5 seeds, final iterates, symmetric noise.

Expected cost with the Bayes-risk rule (temperature-scaled probabilities; assumed cost matrix with in-group cost c_{in}):

c_{in}	cost argmax	cost Bayes	error argmax	error Bayes
0.1	0.0570	0.0551	16.0%	16.1%
0.3	0.0798	0.0788	16.0%	16.1%
0.6	0.1139	0.1131	16.0%	16.0%

2.4 Discussion

Findings and implications

- **Task performance.** 15.95% test error (CI [15.14%, 16.66%]) vs 90 % chance and 32 % for a no-training linear baseline; macro-F1 0.839. Errors are systematic: the upper-body classes, especially Shirt (F1 0.59).
- **H1 supported (as association).** The plug-in complexity $\|\hat{W}\|_F R/\sqrt{n}$ predicts the gap ($\rho = 0.99$ for CE) although the parameter count is constant; parameter count is not what controls generalisation here. This tests the *shape*, not the bound's constants (which are loose).
- **H2 supported.** CE exaggerates over-fitting (+61% vs +15% for error) and a single scalar T changes test CE 24× with zero change in any decision. A model comparison on CE alone can therefore mislead; we need decision metrics *and* a calibration check. Confident mistakes dominate the loss (5 % of points = 36.1%) while 27.6% of errors are almost free for the loss — the disagreement goes both ways.
- **H3 supported.** With $\lambda = 0$ the optimum is not attained; $\|W\|$ grows like log t ($R^2 = 0.999$). The stopping time is a hidden regulariser, which is why λ must be chosen with converged solvers (the reason for implementing L-BFGS).
- **H4 supported under stated conditions.** 40 % symmetric noise: GCE(0.7) 21.8%–22.7% vs CE 34.9%–40.5%; clean-label cost nil. At λ = 1e-3 GCE is even better on clean data (18.4 % vs 20.1 %), which I attribute to CE over-fitting n = 3000 rather than to GCE being intrinsically better.
- **Cost-aware decisions: small effect.** The Bayes-risk rule lowers the assumed expected cost by ≈ 0–3.5 % and raises 0-1 error by 0.02–0.25 pp: the two objectives genuinely differ, but this model's posteriors are rarely ambiguous in a way that changes many decisions.

- **Optimisation.** $\kappa \approx 10^4$: GD at 1/L is very slow; Adam and especially L-BFGS handle the conditioning. 1/L is a safe, not optimal, step (GD still converges at 8/L).

Limitations

(1) Linear hypothesis space: ≈ 4 pp behind a small MLP — approximation error dominates; (2) single dataset and single train/validation split for the final model (CV covers λ only; the bootstrap CI reflects test-set sampling); (3) H1 is tested via a plug-in proxy with 2 seeds — association, not proof; (4) the cost matrix is assumed; (5) GCE experiments: n = 3000, symmetric noise, final iterates, non-convex objective; (6) the over-fitting demonstration is one small-n regime; (7) std estimates from 5 folds/seeds are themselves noisy; (8) tuning of Adam learning rates was coarse.

Potential improvements / future work

Architecture: random-feature/kernel or convolutional features (texture and sleeve-length cues for Shirt vs T-shirt). **Algorithmic:** (a) a calibration- or cost-aware training objective instead of post-hoc repair; (b) noise-transition-matrix correction for class-dependent label noise (the most confident errors look like ambiguous labels); (c) compare the log t drift against an explicit multi-class hard-margin SVM; (d) PCA whitening / pre-conditioning to reduce κ; (e) estimate the bound's constants through Monte-Carlo empirical Rademacher complexity to test H1 quantitatively.

3. Implementation log

<STUDENT: read, correct and extend this log so it truthfully reflects YOUR process before submitting — the specification penalises text you cannot defend in the Q&A.>

3.1 Challenges and solutions

#	Challenge (what I observed)	Diagnosis and solution
1	Gradient check failed my 1e-6 threshold ($1.2 \cdot 10^{-6}$ relative error for CE).	Suspected a bug, but the error was at the level of finite-difference truncation/round-off (h = 1e-6). With h = 1e-5 errors were $7.7 \cdot 10^{-7}$ (CE) and $1.1 \cdot 10^{-7}$ (GCE) and the independent Hessian-vs-gradient test agreed to $4 \cdot 10^{-11}$, so I relaxed the threshold to 1e-5 and documented why.
2	With λ = 1e-3 my L-BFGS reference and scikit-learn disagreed (objective $2.7 \cdot 10^{-4}$; probabilities up to 0.26); neither hit its tolerance.	Not a correctness bug but ill-conditioning ($\kappa \leq 10^4$–10^5) and non-convergence. Moved the comparison to λ = 1e-2, 1500 iterations; agreement became $6 \cdot 10^{-13}$. Lesson: a "reference optimum" must itself be verified converged ($\|\nabla J\| = 2 \cdot 10^{-7}$).
3	Naive Update beat gradient descent in my first plot.	The comparison was unfair (GD η = 0.01 untuned; equal iterations but 100× the compute for random search). Re-ran with η = 1/L and an x-axis in forward-pass equivalents.
4	My first loss-vs-error "decoupling" experiment (n = 5,000) was weak: val CE rose only ≈ 2.6 %.	Re-ran at n = 2,000 (lr 3e-3) and, importantly, **changed the claim** from "decoupled" to "CE exaggerates over-fitting"; the stronger demonstration is temperature scaling, where the decisions are provably identical.
5	Fitted temperature of the over-fit model sat on my grid boundary (T = 5).	Extended the grid to [0.2, 100]; optimum T = 44.5 (CE 16.1 → 0.66).

6	GCE beat CE even on clean labels at λ = 1e-3, suspicious for a claim about robustness.	Suspected a regularisation confound; ran both losses on a matched λ grid. The clean-data gap shrinks at λ = 1e-2 while the 40 %-noise gap persists, so I report the robustness claim and flag the clean-data difference as over-fitting.
7	Cost-sensitive decisions gave only a small gain.	I did not tune the assumed cost matrix to enlarge the effect; I added a sensitivity analysis over c_{in} and reported the modest result.
8	Numerical stability and runtime: naive softmax overflows; the full notebook took ≈ 15 min.	Log-sum-exp shift (checked with logits × 1000); reduced the MLP epochs, seeds at large n and CV subset size; cached data download with an OpenML fallback.

3.2 Use of AI tools

Tool and role. <STUDENT: name the tool(s) you actually used and when> — in this project an AI coding assistant (Claude Code) was used to: read the A2 specification and map it to the rubric; propose the study design (model, hypotheses, experiments); draft the NumPy implementation, the notebook and this report; and run the experiments. **What I (the student) did:** <STUDENT: describe your own decisions, reading, and edits>.

Critical review of AI output (examples). (1) The AI-derived GCE gradient was *not* trusted until it passed a finite-difference test; (2) the AI's textbook claim "GD is safe at η = 1/L and diverges beyond 2/L" turned out to be loose for this problem (GD still converges at 8/L) — I explain this by the ½ bound on the softmax Hessian being attained only for two-class ties; (3) hypotheses the AI proposed were partly *not* supported (H2 as originally phrased, "decoupling", was weak) and were reformulated rather than kept; (4) all literature citations were produced from memory and must be checked against the originals <STUDENT: tick when verified>; (5) every number in this report is read from `results_opt1.json` produced by the notebook, not typed by hand.

3.3 Knowledge gaps

Component I use but do not fully understand	Why necessary	How verified	Attempts to understand
Constants of the multi-class norm-based Rademacher bound (vector-contraction)	H1 uses the *scaling* BR/√n	Empirical slope/Spearman test (shape only); cross-referenced Kakade et al. (2008)	<STUDENT: e.g. worked through the binary case; not the multi-class constant>
Proof of the implicit-bias theorem (Soudry et al.)	Interprets the log t growth (H3)	Fit ‖W‖ = a ln t + c, R^2 = 0.999; margin increases	<STUDENT: e.g. read the statement and intuition; not the proof>
Derivation of the L-BFGS two-loop recursion from the BFGS secant update	Needed to converge ill-conditioned fits	Quadratic unit test; agreement with scikit-learn to $6 \cdot 10^{-13}$	<STUDENT: e.g. followed Nocedal & Wright ch. 7>
Theory of noise-robustness of bounded losses (Ghosh et al. symmetric condition)	Motivates GCE (H4)	Experiment only (5 seeds, two λ)	<STUDENT: e.g. read the symmetric-loss condition>
Bias of the binned ECE estimator	Calibration metric	Reported with Brier and CE and reliability diagrams, not alone	<STUDENT>

4. Presentation materials (optional)

5-minute plan (≈ 2 min on one aspect I invested in)

Time	Content
0:00–0:40	Open with I/O: 784-dim standardised vector in; probability vector over 10 classes out; compare $\hat{y}$ to the label (0-1) while training with cross-entropy. RQ: how faithful is the surrogate?
0:40–1:40	Model and theory-to-code: show `ce_loss_grad` ($\partial\ell/\partial z = q - e_y$), `make_objective`, and the gradient/Hessian/sklearn verification.
1:40–3:40	**Deep dive:** temperature scaling demo — same decisions (22.83 % error), CE 16.1 → 0.66; why $\|W\| \propto \log t$. Bait: "why can the loss change 24× while the error does not?"
3:40–4:30	Results: 15.95 % test error, Shirt failure, GCE under 40 % noise.
4:30–5:00	Limitations and future work.

Anticipated questions (short answers)

- *Why is $\partial\ell/\partial z = q - e_y$?* $\ell = -z_y + \log \Sigma e^{z_j}$; the derivative of the log-sum-exp is softmax.
- *Why 785 zero eigenvalues of the Hessian?* Shift invariance: adding v to every class column leaves q unchanged, so those directions have zero curvature (verified: d+1 zero eigenvalues on a small problem).
- *Why L-BFGS and not Adam for CV?* I need converged solutions; otherwise the stopping time regularises and confounds λ. Adam is shown for conditioning comparison.
- *Why is 1/L safe but GD works at 8/L?* L uses $\lambda_{max}(\text{diag } q - qq^T) \le ½$, attained only for two-class ties; near the optimum most points are confident so curvature is far smaller.
- *Why does GCE help?* Its gradient is CE's times q_y^q, so points the model finds very unlikely (candidate mislabels) get little weight.
- *Does temperature scaling change accuracy?* No: argmax(z/T) = argmax(z).
- *Is the generalisation bound verified?* No — only its shape (association with a plug-in proxy); constants are unknown/loose.
- *I don't know…* is acceptable: state what you would check (the notebook cell, a reference).

References

Bartlett & Mendelson (2002) JMLR 3. · Bartlett, Jordan & McAuliffe (2006) JASA 101. · Berger, Della Pietra & Della Pietra (1996) Comput. Linguistics 22. · Berkson (1944) JASA 39. · Bridle (1990) Neurocomputing, Springer. · Cox (1958) JRSS-B 20. · Ghosh, Kumar & Sastry (2017) AAAI. · Guo, Pleiss, Sun & Weinberger (2017) ICML. · Jaynes (1957) Phys. Rev. 106. · Kakade, Sridharan & Tewari (2008) NeurIPS. · Kingma & Ba (2015) ICLR. · Liu & Nocedal (1989) Math. Programming 45. · McFadden (1974) Frontiers in Econometrics. · Nocedal & Wright (2006) Numerical Optimization. · Rosenblatt (1958) Psychol. Rev. 65. · Soudry et al. (2018) JMLR 19. · Theil (1969) Int. Econ. Rev. 10. · Xiao, Rasul & Vollgraf (2017) arXiv:1708.07747. · Zhang (2004) Ann. Stat. 32. · Zhang & Sabuncu (2018) NeurIPS. *(Bibliographic details written from memory — verify before submission.)*